Artifact 16 – Identifying Functions

Driving Question: What makes a relation a function, and how do you check it in a set of ordered pairs, a table, a mapping, or a graph?

Example 1

A snack machine has six buttons. The table shows the price, in dollars, of the item behind each button.

Button	1	2	3	4	5	6
Price	1.25	1.50	1.25	2.00	1.50	1.75

Write the relation as ordered pairs, (button, price).

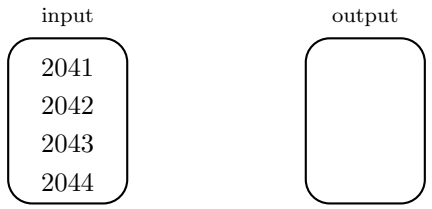
Two buttons cost 1.25 and two cost 1.50. Does that break anything? Explain.

Is the price a function of the button? Explain.

Read it backwards, (price, button). Is the button a function of the price? Explain.

Example 2

Machine Card 5 from your board: a credit union's four accounts and their balances, (account number, balance in dollars). Draw it as a mapping.



(account, balance)

Is the balance a function of the account number? Explain.

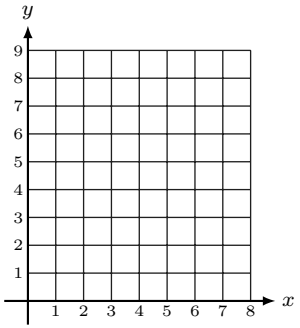
In a mapping, what does "not a function" look like?

Now reverse it, (balance, account). Sketch that mapping on the left. Is the account a function of the balance? Explain.

Example 3

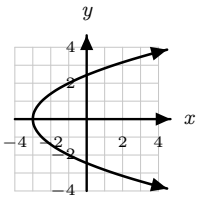
The table shows a relation. Plot its points.

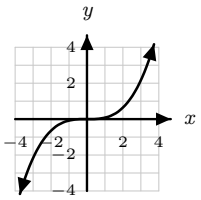
x	1	2	2	5	7
y	3	8	1	6	4



Is y a function of x? Where does the picture show it?

Draw a vertical line through each graph where it would cross the most points. Which graph represents a function? Explain.





Notes to Forgetful Self

Given	Where a repeated INPUT shows	A repeated OUTPUT means
ordered pairs		
a table		
a mapping		
a graph		

A relation is a function when every _____ has exactly one _____.

Does the order of the pairs matter? Does the direction, (x, y) or (y, x), matter?